

# Engineering reference: governed manufacturing reporting

A sanitized technical case study covering architecture, contracts, state, validation, failure handling, tradeoffs, and impact methodology.

## Reading guide

| QUESTION                             | SECTION                            |
|--------------------------------------|------------------------------------|
| How is the platform used?            | 1. Operational contract            |
| What is implemented?                 | 2. Architecture and data model     |
| Why is delivery reliable?            | 3. State, validation, and recovery |
| How is evidence tested?              | 4. Verification matrix             |
| What else was delivered or prepared? | 5. Additional initiatives          |
| How are impact figures qualified?    | 6. Impact methodology              |

## Public/private boundary

This reference uses synthetic data and sanitized topology. Tenant identifiers, employee data, production URLs, credentials, exact commands, permissions, and successor procedures remain in the private handoff.

# 1. Operational contract

The platform prepares decision-support evidence; people retain operational authority.

| OUTPUT            | PRIMARY USER           | PLATFORM RESPONSIBILITY                                     | HUMAN RESPONSIBILITY                             |
|-------------------|------------------------|-------------------------------------------------------------|--------------------------------------------------|
| Shift Draft       | Supervisor             | Collect, reconcile, organize, prepare spreadsheet           | Review exceptions, add context, finalize         |
| Daily Production  | Manager                | Combine production, molding, labor, quality, plan, downtime | Investigate constraints and prioritize follow-up |
| Weekly Production | Managers / teams       | Create comparable period and trend views                    | Plan capacity and labor; prioritize improvement  |
| Weekly Downtime   | Managers / maintenance | Classify recurring and isolated downtime evidence           | Set maintenance and coordination priorities      |

## Decision boundary

Automation may prepare, classify, validate, refuse, and report state. It does not autonomously change staffing, production plans, maintenance priorities, or cross-department actions.

## 2. Architecture and data model

| LAYER                  | COMPONENTS                                              | RESPONSIBILITY                                                                  |
|------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------|
| Operational inputs     | Email, network files, Excel source reports              | Preserve the evidence people already produce                                    |
| Reporting core         | Python ingestion/reconciliation, PostgreSQL             | Normalize evidence and maintain source-linked serving state                     |
| Controlled outputs     | Workbooks and SharePoint delivery                       | Expose prepared shift, daily, weekly, and downtime reports                      |
| Additional initiatives | Forms/QR, machine events, notifications, SPFx dashboard | Extend intake and machine-status visibility without changing decision authority |

### Eight source-linked reporting tables

The serving model separates hourly production, quality, labor, molding, machine notes/events, final reports, report state, and lineage/settlement evidence. Separation prevents an output-friendly aggregate from erasing source identity or missingness.

| CONTRACT                   | WHY EXPLICIT                                                   |
|----------------------------|----------------------------------------------------------------|
| Report identity            | Prevents one period/family from settling another               |
| Material identity          | Distinguishes governed fact changes from capture-only metadata |
| Source lineage             | Connects each prepared value to qualified evidence             |
| Semantic certification     | Tests meaning rather than file appearance alone                |
| Publication/readback state | Separates accepted write from verified remote content          |

# 3. State, validation, and recovery

|           |           |           |           |         |
|-----------|-----------|-----------|-----------|---------|
| CANDIDATE | VALIDATED | PUBLISHED | READ BACK | SETTLED |
|-----------|-----------|-----------|-----------|---------|

State advances only when the evidence required for that transition exists. Missing evidence remains missing; an observed zero remains zero. A successful request is not equivalent to a verified remote artifact.

## Failure scenarios

| SCENARIO                      | CLASSIFICATION   | SAFE BEHAVIOR                                      | AUTHORITY           |
|-------------------------------|------------------|----------------------------------------------------|---------------------|
| Required source absent        | pending_source   | Retain expected output; do not publish             | No write            |
| Equivalent replay             | exact_no_op      | Conserve material identity and remote version      | No write            |
| Write accepted; response lost | readback_pending | GET exact item; never repeat PUT                   | GET only            |
| Contract or lineage drift     | contract_refusal | Stop interpretation/publication                    | No write            |
| Eligible late source          | bounded_recovery | Advance same output identity once after validation | One qualified write |

## Release and rollback

One-writer releases serialize ownership and prevent overlapping publication. Promotion is guarded by output-suppressed/current-data checks. Exact rollback restores the last known-good immutable revision rather than reconstructing a deployment from mutable state.

## 4. Verification matrix

| LAYER                | REPRESENTATIVE CHECK                                     | FAILURE PREVENTED                     |
|----------------------|----------------------------------------------------------|---------------------------------------|
| Source facts         | Conservation and reconciliation against qualified inputs | Dropped or duplicated evidence        |
| Workbook package     | Visible sheets, formulas, structure, stored values       | Client-dependent or malformed output  |
| Semantics            | Missing-versus-zero and family-specific rules            | Meaning drift                         |
| Normalized rendering | Stable content independent of packaging noise            | False changes from metadata           |
| Remote readback      | Exact item, version, ETag, content/material hash         | Accepted-write ambiguity              |
| Replay               | Immediate equivalent cycle makes zero second PUT         | Duplicate versions                    |
| Release              | Serialized handoff, readiness guard, exact rollback      | Concurrent writers / unsafe promotion |

### Claim discipline

Validation coverage proves the tested invariants and scenarios. It is not presented as blanket accuracy. Public evidence is sanitized; production-specific receipts and exact successor tests remain private.

### Selected tradeoffs

| CHOICE                | BENEFIT                                | COST                                        |
|-----------------------|----------------------------------------|---------------------------------------------|
| Fail closed           | No silent advancement on ambiguity     | Some reports settle later                   |
| One writer            | Clear ownership and recovery semantics | Serialized promotion                        |
| Same-item advancement | Stable operational identity and links  | Requires strict replay/material checks      |
| Bounded repair        | Recovers only evidence-supported cases | Some historical cases remain terminal holds |

## 5. Additional initiatives

| INITIATIVE               | STATUS                         | PURPOSE                                                                                                 | BOUNDARY                                                         |
|--------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Machine-event visibility | Delivered extension            | QR/Form intake, notifications, and SharePoint dashboard status                                          | Visibility only; people investigate and decide                   |
| Reporting Assistant      | Functional read-only extension | Answers bounded queries with source-grounded retrieval, citations, diagnostics, validation, and refusal | No write authority; broader model-backed/advisory work is future |

### Technical foundation

The assistant separates bounded retrieval, read-only report-doctor diagnostics, and governed-oracle validation because they have different authority and refusal behavior. Those functional paths reuse governed identity, lineage, state, and semantic contracts; they do not create a new writer.

### Human control

Machine visibility and bounded retrieval can shorten the path to evidence. Supervisors and managers still interpret context and choose operational action.

## 6. Impact methodology

| VALUE AREA          | BASIS                                                                                                       | RESULT                                                           | QUALIFICATION                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Core reporting      | 20 supervisor + 10 manager hours/week; 52 weeks; documented role-rate assumptions                           | 30 hours/week; 1,560 hours/year; about \$52K                     | Modeled labor-capacity value, not booked savings                               |
| Assistant retrieval | Representative 20-30 minute search compared with a 1-2 minute cited response                                | About 90% faster                                                 | Estimated from the functional assistant; not a controlled production benchmark |
| Machine visibility  | 5-minute response improvement; 1,000 production units/hour/machine; 52.5 events/week; 52 weeks; \$0.90/unit | Up to 227,500 production units; about \$205K contribution margin | Additional-initiative upper bound, not realized production or margin           |

### Limitations

The labor, assistant-retrieval, and machine-visibility figures represent separate value pools and are not additive. The models depend on representative time studies, event frequency, throughput, response improvement, role rates, and contribution margin. They support prioritization, not guarantees. The synthetic workbook separates demonstrated controls from documented impact assumptions.

### Implementation references

The public repository contains sanitized architecture, state-machine documentation, tests, synthetic scenarios, release chronology, and claim boundaries. The private successor package contains exact dependencies, permissions, operating procedures, rollback, credentials transfer, and acceptance checks.